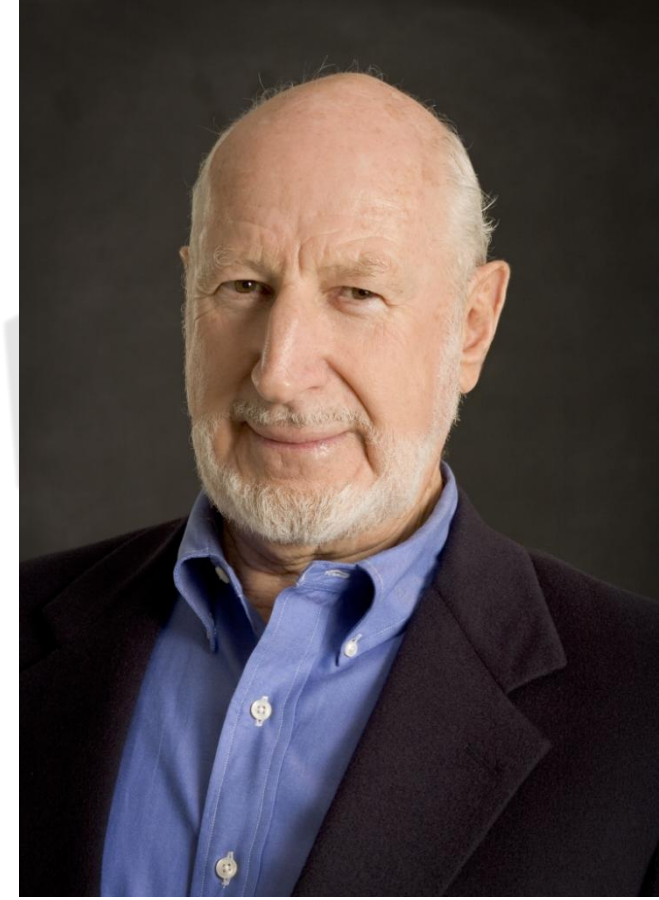


DLL: Access Control

ALOHA Protocols

Aloha

- Earliest random-access method, was developed at the University of Hawaii around 1970.
- It was designed for a radio (wireless) LAN, but it can be used on any shared medium.
- The original ALOHA protocol is called pure ALOHA. This is a simple, but elegant protocol.
- The idea is that each station sends a frame whenever it has a frame to send. However, there is the possibility of collision between frames from different stations.

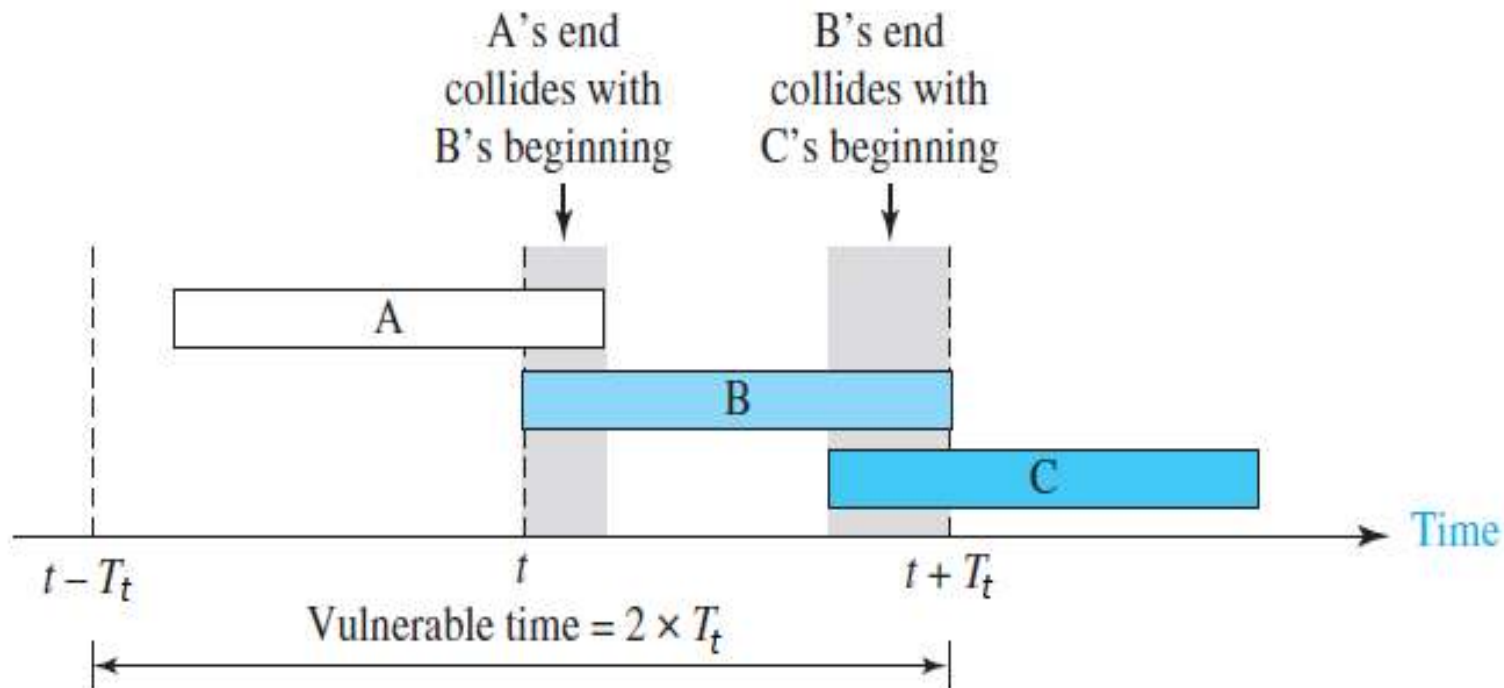


Norman Manuel Abramson

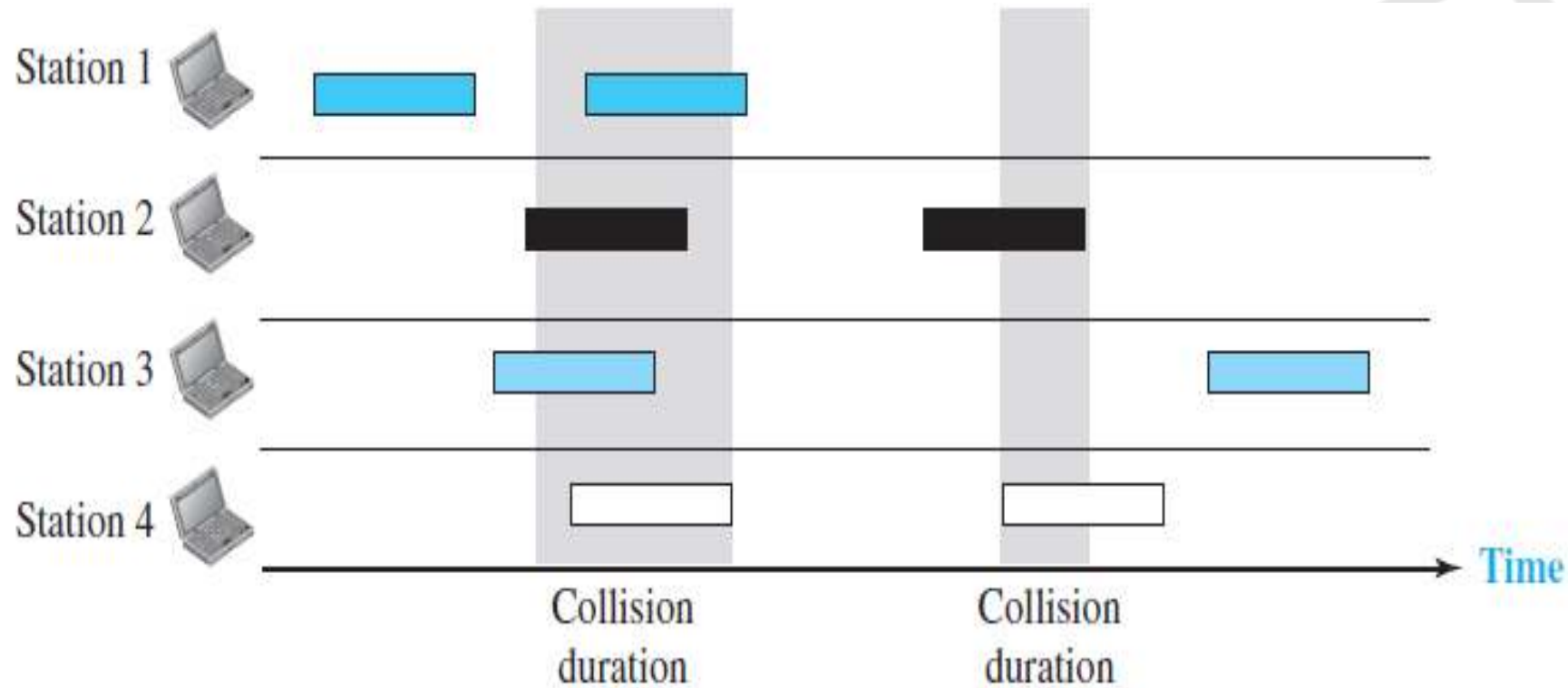
- **Propagation Delay**: Propagation delay is the time it takes for a bit to travel from point A to point B in the transmission media.
 $T_p = (\text{Distance}) / (\text{Propagation speed})$

- **Transmission Delay (TT)** : A sender needs to put the bits in a packet on the line one by one. If the first bit of the packet is put on the line at time t_1 and the last bit is put on the line at time t_2 , transmission delay of the packet is $(t_2 - t_1)$.
 $T_t = (\text{Packet length (L)}) / (\text{Transmission rate or Bandwidth (B)}) = L / B$

- Vulnerable time in which there is a possibility of collision. We assume that the stations send fixed-length frames with each frame taking T_{fr} S to send.
- Station A sends a frame at time t . Now imagine station B has already sent a frame between $t - T_{fr}$ and t . This leads to a collision between the frames from station A and station B. The end of B's frame collides with the beginning of A's frame.
- On the other hand, suppose that station C sends a frame between t and $t + T_{fr}$. Here, there is a collision between frames from station A and station C. The beginning of C's frame collides with the end of A's frame. we see that the vulnerable time, during which a collision may occur in pure ALOHA, is 2 times the frame transmission time.
- Pure ALOHA vulnerable time = $2 \times T_{fr}$



- The pure ALOHA protocol relies on acknowledgments from the receiver. When a station sends a frame, it expects the receiver to send an acknowledgment.
- If the acknowledgment does not arrive in time-out period, the station assumes that the frame (or the acknowledgment) has been destroyed and resends the frame.



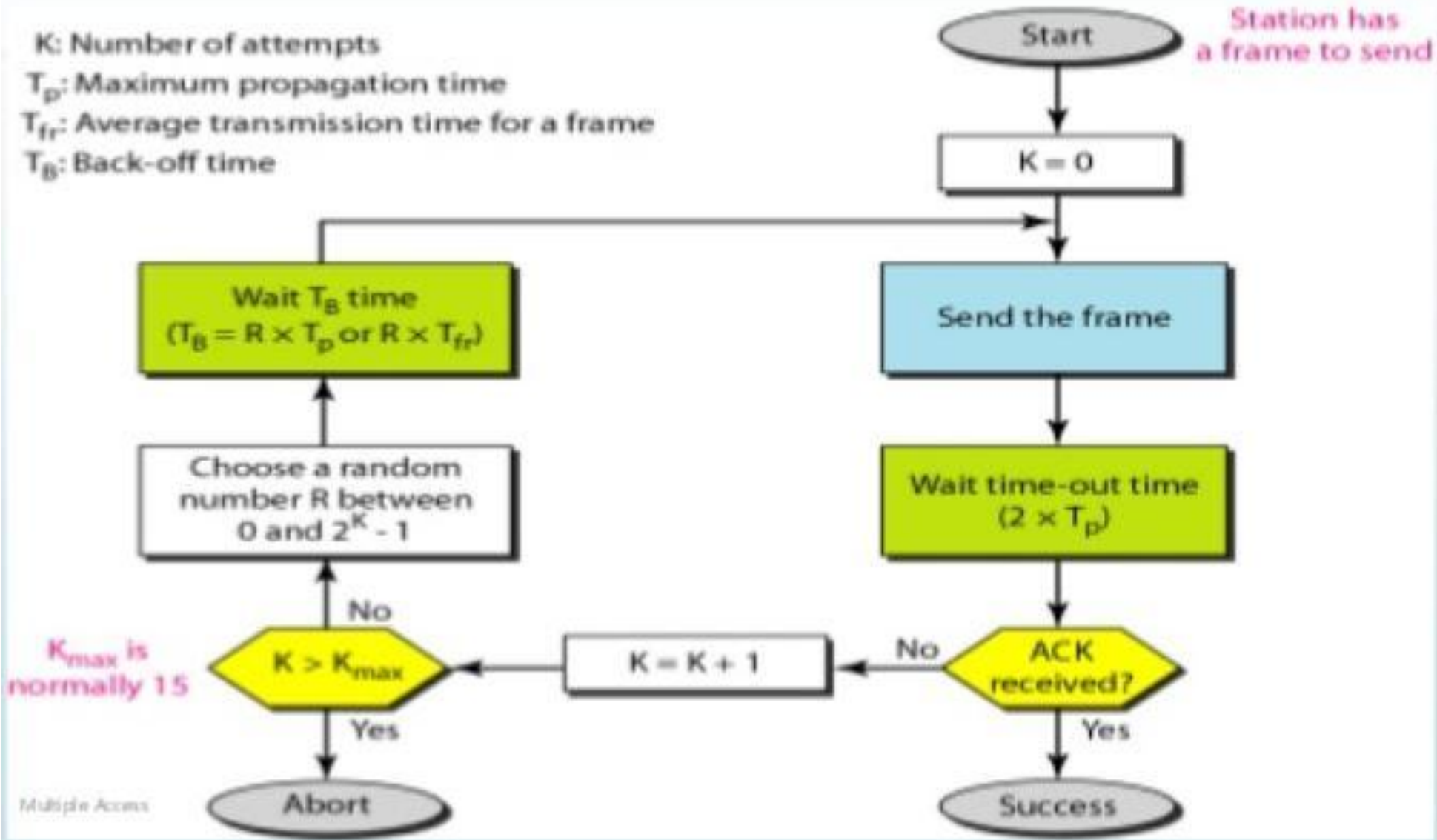
Procedure for Pure ALOHA protocol

K: Number of attempts

T_p : Maximum propagation time

T_{fr} : Average transmission time for a frame

T_B : Back-off time



- If all these stations try to resend their frames after the time-out, the frames will collide again.
- Pure ALOHA dictates that when the time-out period passes, each station waits a random amount of time before resending its frame. The randomness will help avoid more collisions. We call this time the back-off time T_B .
- Pure ALOHA has a second method to prevent congesting the channel with retransmitted frames. After a maximum number of retransmissions attempts $K_{\max}$ a station must give up and try later.

Example: The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s . Find back off time possibility after two consecutive collision ?

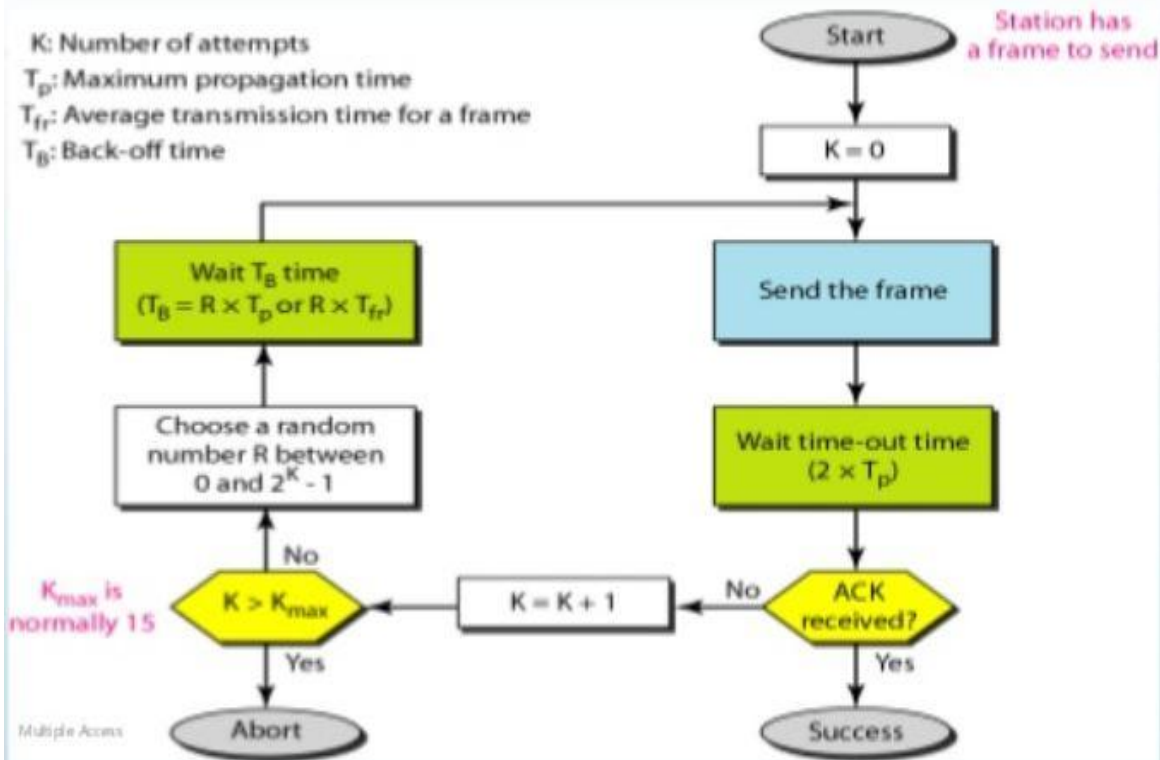
Procedure for Pure ALOHA protocol

K: Number of attempts

T_p : Maximum propagation time

T_{fr} : Average transmission time for a frame

T_B : Back-off time



Example 12.3

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

- a. 1000 frames per second?
- b. 500 frames per second?
- c. 250 frames per second?

Solution

The frame transmission time is $200/200$ kbps or 1 ms.

- a. If the system creates 1000 frames per second, or 1 frame per millisecond, then $G = 1$. In this case $S = G \times e^{-2G} = 0.135$ (13.5 percent). This means that the throughput is $1000 \times 0.135 = 135$ frames. Only 135 frames out of 1000 will probably survive.
- b. If the system creates 500 frames per second, or $1/2$ frames per millisecond, then $G = 1/2$. In this case $S = G \times e^{-2G} = 0.184$ (18.4 percent). This means that the throughput is $500 \times 0.184 = 92$ and that only 92 frames out of 500 will probably survive. Note that this is the *maximum* throughput case, percentage-wise.
- c. If the system creates 250 frames per second, or $1/4$ frames per millisecond, then $G = 1/4$. In this case $S = G \times e^{-2G} = 0.152$ (15.2 percent). This means that the throughput is $250 \times 0.152 = 38$. Only 38 frames out of 250 will probably survive.

Example: A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Example 12.2

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Solution

Average frame transmission time T_{fr} is 200 bits/200 kbps or 1 ms. The vulnerable time is $2 \times 1 \text{ ms} = 2 \text{ ms}$. This means no station should send later than 1 ms before this station starts transmission and no station should start sending during the period (1 ms) that this station is sending.

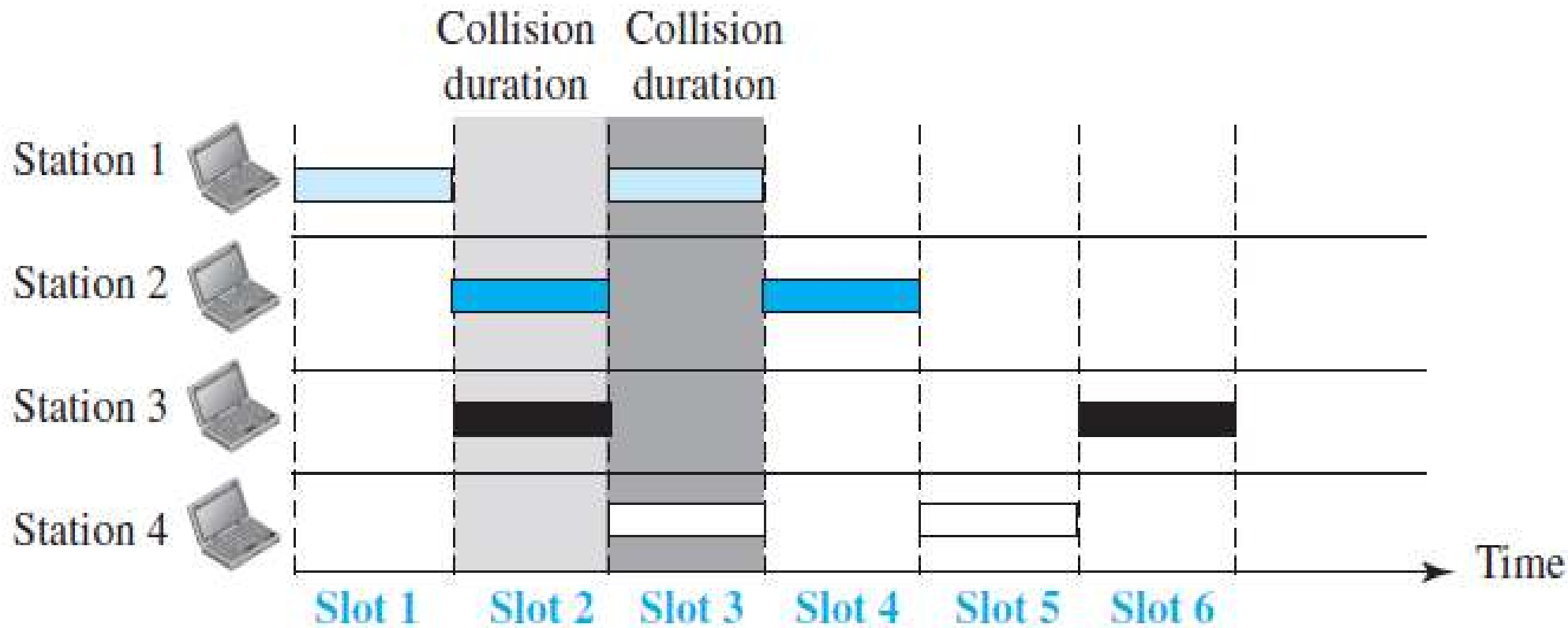
Throughput

Let us call G the average number of frames generated by the system during one frame transmission time. Then it can be proven that the average number of successfully transmitted frames for pure ALOHA is $S = G \times e^{-2G}$. The maximum throughput S_{max} is 0.184, for $G = 1/2$. (We can find it by setting the derivative of S with respect to G to 0; see Exercises.) In other words, if one-half a frame is generated during one frame transmission time (one frame during two frame transmission times), then 18.4 percent of these frames reach their destination successfully. We expect $G = 1/2$ to produce the maximum throughput because the vulnerable time is 2 times the frame transmission time. Therefore, if a station generates only one frame in this vulnerable time (and no other stations generate a frame during this time), the frame will reach its destination successfully.

**The throughput for pure ALOHA is $S = G \times e^{-2G}$.
The maximum throughput $S_{max} = 1/(2e) = 0.184$ when $G = (1/2)$.**

Slotted ALOHA

- Pure ALOHA has a vulnerable time of $2 \times T_{fr}$. This is so because there is no rule that defines when the station can send. A station may send soon after another station has started or soon before another station has finished.
- Slotted ALOHA was invented to improve the efficiency of pure ALOHA. In slotted ALOHA we divide the time into slots of T_{fr} and force the station to send only at the beginning of the time slot.



- Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next time slot. This means that the station which started at the beginning of this slot has already finished sending its frame.
- Off course, there is still the possibility of collision if two stations try to send at the beginning of the same time slot. However, the vulnerable time is now reduced to one-half, equal to T_{fr}

